

REMARKS

Applicants respectfully request reconsideration of this patent application, particularly in view of the above Amendment and the following remarks. Claims 1-2, 7, and 12 are currently pending, and claims 3-6, 8-11, and 13-22 have been withdrawn.

Amendments to the Specification

The Examiner objected to claims 1, 7, and 12 as informal because the claims recited “*Terminalia bellerica*,” which the Examiner stated is considered a commonly used misnomer in the art. Paragraph [0006] is corrected to state “*Terminalia bellirica* (Gaertn.) Roxb. (a.k.a. as *Terminalia bellerica*) (“TB”).”

Amendments to the Claims

Claims 1, 5-7, 10-12, and 18-20 are amended to delete “*bellerica*” and recite “*bellirica*.” Claims 1, 5-7, 10-12, and 18-20 are further amended to recite that the *Terminalia bellirica* extract is an aqueous extraction. Support for this amendment is found, for example, on page 18, at paragraph [0086].

Claims 1 and 7 are further amended to delete “comprising” and recite “consisting essentially of.” Claim 12 is amended to recite “consisting essentially of.”

Claims 1, 7, and 12 are amended to recite “wherein the aqueous extract comprises at least 15% based on the total weight of the extract of one or more bioactive low molecular weight hydrolyzable tannins selected from the group consisting of chebulinic acid and chebulagic acid.” Support for this amendment is found, for example, on page 10, at paragraph [0045], and page 18, at paragraph [0086].

No new matter has been added by this Amendment. Applicants believe that no fees are owed because the number of claims currently pending does not exceed the number originally paid for. However, if any additional fees are required (to maintain the pendency of this application or to timely enter this responsive paper) or an overpayment of fees is made, the Commissioner is hereby authorized to debit or credit our Deposit Account No. 50-3394, as necessary.

Examiner Interview of April 13, 2017

Applicants thank the Examiner for the telephonic Interview on April 13, 2017. Applicants agreed to amend the specification at the first instance of the objected term “*bellerica*”

so as to overcome the specification objection. Applicants thank the Examiner for the discussion regarding suggested amendments to overcome the maintained rejections.

Examiner Interview of April 20, 2017

Applicants thank the Examiner for the telephonic Interview of April 20, 2017, consideration of the Proposed Amendment, which is reflected in the above Listing of Claims, and suggestion for requesting consideration under the After Final Consideration Program 2.0.

Claim Objections

The Examiner objected to claims 1, 7, and 12 as informal because the claims recited “*Terminalia bellerica*,” which the Examiner stated is considered a commonly used misnomer in the art. As stated above in the Amendments to the Claims, claims 1, 5-7, 10-12, and 18-20 have been amended to delete “*bellerica*” and recite “*bellirica*.” Accordingly, Applicants respectfully request withdrawal of the Examiner’s objection.

Claim Rejections – 35 U.S.C. § 102

Rejection of Claims 1-2 and 12

The rejection of claims 1-2 and 12 under 35 U.S.C. § 102(a)(1) as being anticipated by Caraka Samhita (Sharma (Ed. & transl), vol I-II: Chaukhamba Orientalia, Varanasi, 5th Edn. 2000. [Time of origin 1000 BC-4th century] (Trad. Know. Dig. Lib. (TKDL), 1 pg.; herein “Caraka”) as evidenced by Wikipedia: Hyperuricemia (Online, URL: <<https://web.archive.org/web/20120203161708/http://en.wikipedia.org/wiki/Hyperuricemia>>, 3 Feb 2012, 3 pp., herein “Wiki (I)”) is respectfully traversed.

The Examiner stated in the 3/8/17 Final Office Action, “Caraka discloses a decoction (water extract which is boiled and filtered to obtain a kvatha, (Ayurvedic medicine a.k.a.: quatha, kwatha), of *T. chebula* (TC) and *T. bellirica* (TB) fruit along with *Phyllanthus embilica* fruit as a treatment for gout.” (See, e.g., 3/8/17 Final Office Action, at 3) (emphasis in original). As reflected above in the Amendments to the Claims, the Applicants have amended claims 1 and 12 to recite that the composition administered according to the recited methods **consists essentially of Terminalia bellirica** aqueous extract instead of **comprises Terminalia bellirica** aqueous extract. Caraka does not disclose that *T. bellirica* aqueous extract is the active ingredient of the disclosed decoction, and as Dr. Sanyasi Kalidindi stated in his Declaration of October 23, 2016, submitted as Appendix A to the October 24, 2016 Amendment in the instant case, it is highly likely that only *Phyllanthus embilica* extract would survive boiling temperatures such as to retain

any efficacy for a treatment of any disease or condition, whether the disease or condition be gout or otherwise. (See, e.g., Appendix A to 10/24/16 Amendment, at ¶ 22).

Therefore, because claims 1-2 and 12, as amended, do not recite an aqueous extract of *T. chebula* (TC) fruit or *Phyllanthus embilica* fruit as a treatment for gout, claims 1-2 and 12 are not anticipated by Caraka.

For at least the reasons above, Applicants respectfully submit that Claims 1-2 and 12 are not anticipated by Caraka. Accordingly, reconsideration and withdrawal of this rejection is respectfully requested.

Rejection of Claim 7

The rejection of claim 17 under 35 U.S.C. § 102(a)(1) as being anticipated by Naik et al. (“Free Radical Scavenging Reactions and Phytochemical Analysis of Triphala, an Ayurvedic Formulation”; Current Science, 25 April 2006, 90(8), 1100-1105) (“Naik et al.”) is respectfully traversed.

The Examiner stated in the 3/8/17 Final Office Action,

Naik et al. teach that Triphala is an Ayurvedic formulation made by extracting fruits of ***three plants, one of the fruits originating from TB fruit*** and whereby the fruits are extracted with water (see protocol for making Triphala: col. 2, page 1100). . . . Therefore, claim 7 is anticipated by Naik et al. because xanthine oxidase would have necessarily been inhibited in the persons ***taking Triphala*** prior to the document published by Naik et al. as evidenced by Naik et al. Please note that claim 7 is directed toward any inhibition amount of XO and, therefore, administration of ***Triphala*** need have only inhibited one molecule of XO. Since ***Triphala*** (water extract thereof) has XO inhibiting capabilities, ***this composition*** would have inhibited at least one mole of XO upon *in-vivo* administration.

(See, e.g., 3/8/17 Final Office Action, at 9) (emphases added). As reflected above in the Amendments to the Claims, the Applicants have amended claim 7 to recite that the composition administered according to the recited methods **consists essentially of** *Terminalia bellirica* aqueous extract instead of **comprises** *Terminalia bellirica* aqueous extract. Naik et al. does not disclose that *T. bellirica* aqueous extract is the active ingredient of Triphala.

Therefore, because claim 7, as amended, does not recite an Ayurvedic formulation made by extracting fruits of three plants, one of the fruits originating from TB fruit, claim 7 is not anticipated by Naik et al.

For at least the reasons above, Applicants respectfully submit that Claim 7 is not anticipated by Naik et al. Accordingly, reconsideration and withdrawal of this rejection is respectfully requested.

Conclusion

Applicants respectfully submit that the subject application is in condition for allowance. If the Examiner feels that any issues remain regarding this application, then Applicants' undersigned representative would like to discuss the case with the Examiner. The undersigned can be reached at (312)-784-1051.

Respectfully submitted,

Dated: April 28, 2017

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THE UNITED STATES PATENT AND TRADEMARK OFFICE

Applicants:	KALIDINDI, Sanyasi R. et al.		
Application No.:	14/459,046	Group No.:	1655
Filing Date:	13 August 2014	Examiner:	Kosar, Aaron J.
Customer No.:	51079	Confirmation No.:	2412
Title:	TERMINALIA CHEBULA AND TERMINALIA BELLERICA EXTRACTS FOR INHIBITION OF XANTHINE OXIDASE		

AMENDMENT AND RESPONSE TO OFFICE ACTION

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Commissioner for Patents
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Alexandria, VA 22313-1450

In response to the Final Office Action mailed 8 March 2017, Applicants respectfully request entry of the claims and consideration of the following remarks.

Amendments to the Specification begin on page 2 of this paper.

Amendments to the Claims are reflected in the Listing of Claims that begins on page 3 of this paper.

Remarks/Arguments begin on page 6 of this paper.

I hereby certify that this correspondence (along with any paper referred to as being attached or enclosed) is being electronically deposited with the United States Patent and Trademark Office addressed to: Mail Stop AF, Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450 on April 28, 2017.

April 28, 2017
Date

/Adam D. Sussman/
Signature

Amendments to the Claims:

This listing will replace all prior versions, and listing, of the Claims in the application. The text of all claims (including withdrawn claims), is set forth below. Cancelled and not entered claims are indicated with claim number and status only. The claims as listed below show added text with underlining and deleted text with ~~[[double brackets]]~~ or ~~strikethrough~~. The status of each claim is indicated with one of (Original), (Currently Amended), (Cancelled), (Withdrawn), (New), (Previously Presented), (Not Entered), or (Withdrawn – Currently Amended).

Listing of Claims:

1. (Currently Amended) A method of treating a free radical-induced disease in an individual, comprising administering to the individual in need of such treatment a therapeutically effective amount of a composition comprising consisting essentially of *Terminalia belleriea bellirica* aqueous extract, wherein the aqueous extract comprises at least 15% based on the total weight of the extract of one or more bioactive low molecular weight hydrolyzable tannins selected from the group consisting of chebulinic acid and chebulagic acid, and wherein xanthine oxidase activity is inhibited.
2. (Previously Presented) The method of claim 1, wherein the free radical-induced disease is selected from the group consisting of gout, uricemia, and hyperuricemia.
3. (Withdrawn) The method of claim 2, wherein the *Terminalia chebula* extract composition comprises not less than about 12% by weight chebulinic acid based on the total weight of the extract, not less than about 10% chebulagic acid based on the total weight of the extract, and not less than about 10% by weight other low molecular weight hydrolyzable tannoids based on the total weight of the extract.
4. (Withdrawn) The method of claim 3, wherein the *Terminalia chebula* extract composition has an effective dose (ED₅₀) of about 6 µg/ml for inhibition of xanthine oxidase.
5. (Withdrawn – Currently Amended) The method of claim 2, wherein the *Terminalia belleriea-bellirica* aqueous extract composition comprises not less than about 2% by weight of chebulinic acid based on the total weight of the extract, not less than about 2% by weight of chebulagic acid based on the total weight of the extract, and not less than about 10% by weight other low molecular weight hydrolyzable tannoids based on the total weight of the extract.

6. (Withdrawn – Currently Amended) The method of claim 5, wherein the *Terminalia bellerica-bellirica* aqueous extract composition has an effective dose (ED₅₀) of about µg/ml for inhibition of xanthine oxidase.

7. (Currently Amended) A method of inhibiting xanthine oxidase in a human individual, comprising administering to the human individual in need of such treatment a therapeutically effective amount of a composition ~~comprising~~ consisting essentially of *Terminalia bellerica-bellirica* aqueous extract, wherein the aqueous extract comprises at least 15% based on the total weight of the extract of one or more bioactive low molecular weight hydrolyzable tannins selected from the group consisting of chebulinic acid and chebulagic acid.

8. (Withdrawn) The method of claim 7, wherein the *Terminalia chebula* extract composition comprises not less than about 12% by weight chebulinic acid based on the total weight of the extract, not less than about 10% chebulagic acid based on the total weight of the extract, and not less than about 10% by weight other lower molecular weight hydrolyzable tannoids based on the total weight of the extract.

9. (Withdrawn) The method of claim 8, wherein the *Terminalia chebula* extract composition has an effective dose (ED₅₀) of about 6 µg/ml.

10. (Withdrawn – Currently Amended) The method of claim 7, wherein the *Terminalia bellerica-bellirica* aqueous extract composition comprises not less than about 2% by weight of chebulinic acid based on the total weight of the extract, not less than about 2% by weight of chebulagic acid based on the total weight of the extract, and not less than about 10% by weight other low molecular weight hydrolyzable tannoids based on the total weight of the extract.

11. (Withdrawn – Currently Amended) The method of claim 10, wherein the aqueous extract composition comprises *Terminalia bellerica-bellirica* having an effective dose (ED₅₀) of about 5 µg/ml.

12. (Currently Amended) A method for treating ~~or preventing~~ hyperuricemia in a human individual, comprising orally administering to the human individual in need of such treatment a therapeutically effective amount of a composition consisting essentially of *Terminalia bellerica-bellirica* aqueous extract, wherein the aqueous extract comprises at least 15% based on the total weight of the extract of one or more bioactive low molecular weight

hydrolyzable tannins selected from the group consisting of chebulinic acid and chebulagic acid, and wherein serum uric acid is lowered.

13. (Withdrawn) The method of claim **12**, wherein the *Terminalia chebula* extract composition comprises not less than about 12% by weight chebulinic acid based on the total weight of the extract, not less than about 10% chebulagic acid based on the total weight of the extract, and not less than about 10% by weight other low molecular weight hydrolyzable tannoids based on the total weight of the extract.

14. (Withdrawn) The method of claim **13**, wherein the *Terminalia chebula* extract composition is provided in a daily dosage range of from about 250 mg to about 2000 mg.

15. (Withdrawn) The method of claim **13**, wherein the *Terminalia chebula* extract composition is provided in a daily dosage range of from about 500 mg to about 1000 mg.

16. (Withdrawn) The method of claim **15**, wherein serum uric acid is lowered to less than about 6.8 mg/dl.

17. (Withdrawn) The method of claim **15**, wherein serum uric acid is lowered to less than about 6.0 mg/dl.

18. (Withdrawn – Currently Amended) The method of claim **12**, wherein the *Terminalia bellerica*-*bellirica* aqueous extract composition comprises not less than about 2% by weight of chebulinic acid based on the total weight of the extract, not less than about 2% by weight of chebulagic acid based on the total weight of the extract, and not less than about 10% by weight other low molecular weight hydrolyzable tannoids based on the total weight of the extract.

19. (Withdrawn – Currently Amended) The method of claim **18**, wherein the *Terminalia bellerica*-*bellirica* aqueous extract composition is provided in a daily dosage range of from about 250 mg to about 2000 mg.

20. (Withdrawn – Currently Amended) The method of claim **18**, wherein the *Terminalia bellerica*-*bellirica* aqueous extract composition is provided in a daily dosage range of from about 500 mg to about 1000 mg.

21. (Withdrawn) The method of claim **20**, wherein serum uric acid is lowered to less than about 6.8 mg/dl.

22. (Withdrawn) The method of claim **20**, wherein serum uric acid is lowered to less than about 6.0 mg/dl.